

Desktop virtualisation method	Test no.	Test case	CPU performance percentage	Load average	Memory used in GB
USB	1	Host PC (server) without any client	1-3 per cent	0.2	0.9
	2	With single client viewing HD video on VLC	20-25 per cent	1.9	2.4
	3	With two clients viewing HD video on VLC	45-50 per cent	2.3	3.1
	4	With three clients viewing HD video on VLC	95 per cent	4.0	4.9
PCI	1	Host PC (server) without any client	1-3 per cent	0.2	0.9
	2	With single client viewing HD video on VLC	14 per cent	0.9	1.1
	3	With two clients viewing HD video on VLC	24 per cent	1.4	1.15
	4	With three clients viewing HD video on VLC	35-36 per cent	1.7	1.3

Table 1: Desktop virtualisation test results—USB vs PCI

This makes the solution even more compelling from the cost perspective.

Desktop virtualisation methods

To explore further, we decided to try out two popular methods of desktop virtualisation—USB and PCI. To evaluate how both of these methods fare, we performed a test using the available USB and PCI solutions to assess the difference in client PC performance, where server configuration remained the same (please refer to the box for server configuration details). We installed Ubuntu 10.04 on the server, then we installed a desktop virtualisation software that is required to connect client machines with the server. We also installed VLC Player. After configuring the server, we connected a total of three clients, first using USB and then PCI.

We then recorded readings for server performance while playing HD video on VLC—initially on the first client; then we connected the second client and played the same video on both client machines. Subsequently, we recorded the performance of the server with three clients connected to it and playing the same HD video.

Here are a few inferences based on the readings as shown in Table 1.

- PCI is technically more reliable than USB for networking. You are less likely to experience dropped connections or a sluggish response time to your modem when using PCI instead of USB.
- Ethernet cables can reach a longer distance than USB cables. A single Ethernet cable can run 10 metres (32 ft) on a direct connection with PCI, while USB cable runs are limited to approximately 5 metres (16 feet).

The test demonstrates that PCI is better suited for network computing, while USB works well for peripheral connectivity.

In a nutshell

According to recent studies, desktop virtualisation solutions result in 75 per cent lower maintenance and 90 per cent lower energy costs. The low entry and reduced lifecycle cost of desktop virtualisation is turning the old economics of PC purchasing and maintenance on its head. Amongst the different variants of virtual desktop devices, PCI scores over USB significantly in terms of performance, optimum utilisation of resources as well as user experience. This indeed makes shared computing using PCI a superior value proposition over USB for organisations that want to enable wider access to computing at a low cost.

Server configuration details

Computer system used: HP MS6000 extended configuration
Operating system: Ubuntu 10.04
Processor: Intel Core 2 Quad Q9500
Chipset: Intel Q43 Express
RAM: 6 GB DDR3 (1333 MHz)
Hard disk drive: 500 GB SATA 3.0 Gb/s
Removable media: 16x SATA DVD writer drive
Graphics: Integrated Intel Graphics Media Accelerator 4500
 Intel Graphics Media Accelerator driver installed using Nvidia graphics driver 96.43.18 for Ubuntu 10.04.

